ANH HOANG PHUC NGUYEN

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PROFILE SUMMARY

I am an AI/ML Engineer and Data Scientist with a Master of Artificial Intelligence and 3 years of experience. I specialise in applying AI solutions to industry products, with hands-on technical execution, particularly in the fields of computer vision and natural language processing.

SKILL

- **Programming Languages:** Python, SQL
- **AI & Data Science:** PyTorch, Tensorflow, LLMs, LoRA, RAG, LangChain, n8n
- **Cloud & Deployment:** AWS (EC2, S3, Bedrock, Lambda), GCP, Docker, CI/CD, OpenRouter, Git
- **Databases:** MySQL, SQLite3, pgvector
- **Others:** Linux, Jira, Tableau, PyQT5, Flask, Apache Kafka, MQTT, RabbitMQ

WORK EXPERIENCE

AI Engineer (Sydney, Australia) — May 2026 - August 2026
Chipforge.ai

- Implemented the Reflexion Loop in the Chipforge platform, fixing HDL bugs in 3-5 iterations.
- Data mining and analysis of around 1600 HDL GitHub repositories, processing the CPT, generating the HDL specifications for SFT data and training Nemotron3-Nano.

Research Assistant (Sydney, Australia) — April 2025 – February 2026
University of Technology Sydney
Project: Scoliosis Assessment Application

- Built data pipeline to process and annotate 20,000 X-ray images for fine-tuning Mask-RCNN.
- Fine-tuned Mask R-CNN models for vertebrae segmentation and proposed spine curve angle measurement method, with an absolute error of $3.02°\pm2.85°$.
- Deployed the system via CI/CD pipeline via GitHub Actions, Docker on AWS EC2, integrating AWS S3 for secure versioning and efficient retrieval of deep learning model weights.

Machine Learning Engineer (Ho Chi Minh City, Vietnam) — April 2023 - January 2024
Ycomm Viet Nam
Project: Webtoon editor – Agile development

- Deployed text inpainting API using LaMa model and text detection APIs using EasyOCR and Google Vision API, which is the main feature in the webtoon editor.
- Built a webtoon dataset pipeline for English, Korean, Japanese, and Chinese using Label Studio.

Project: Subtitle extractor – Agile development

- Deployed Stable_Whisper and Google Speech-to-text API to extract subtitles in cartoons.
- Integrated DeepL API for translation API.

Artificial Intelligence Specialist (Ha Noi and Ho Chi Minh City) — October 2021 - July 2022
VinBigData
Project: Text-to-Speech with subword embedding

- Researched phoneme + subword context modelling based on Tacotron2.
- Developed a custom Vietnamese tokenizer using Hugging Face tokenization tools.
- Achieved Mel Cepstral Distortion (MCD) of 11.37, outperforming phoneme-only (12.1) and subword-only (11.77) models.

Project: Vietnamese Spelling Error Correction System

- Implemented augmentation techniques to create 5 million training samples.
- Experimented with BI-LSTM (ROUGE-1 0.86), and Transformer (ROUGE-1 0.93).

Project: Visual Question Answering

- Implemented pipeline including Faster R-CNN for image feature extraction, GloVe embeddings for question representation, and BERT for context modelling.
- Achieved overall accuracy of 70.98%.

Data Scientist (Ho Chi Minh City, Vietnam) June 2021 - October 2021
FPT Telecom
Project: Unknown Recognition by Camera Project
- Integrated EfficientNet into the backbone of RetinaFace, reducing model weight to 50%.
- Optimized user registration using FAISS, users only provide 3 images to create account.

Artificial Intelligence Intern (Ho Chi Minh City, Vietnam) July 2020 - October 2020
Cloud Nine Solutions
Project: Facial Recognition Check-in Project
- Developed a neural network model for real-time facial recognition.
- Deployed the solution on Jetson Nano for on-device, low-latency check-in.

EDUCATION

Master of Artificial Intelligence - University of Technology Sydney (UTS)
February 2024 - December 2025 **|** GPA: 6.81/7 | WAM: 93.69/100
Postgraduate Academic Excellence International Scholarship
- Participated in Waymo 3D Semantic Segmentation Challenge: trained and validated SpUNet and Point Transformer V3 on Waymo v2.0.1.
- Implemented ClearML for the Job and Candidate recommendation project, nominated for presentation at AI Showcase in UTS FEIT TechFest 2025.

Bachelor of Engineering in Computer Science - Ho Chi Minh City University of Technology (HCMUT)
August 2017 - October 2021 | GPA: 8.52 / 10.00
Certificate of Honour - the Faculty of Computer Science and Engineering.
Thesis: Biometric-Based Attendance Tracking System with One/Fewshot Learning
- Implemented Kalman filter for optimised face tracking and Apache Kafka to connect Jetson Nano and Google Cloud Platform.
- Fine-tuned FaceNet on HCMUT student dataset, achieving F1 score of 99.395% (0.335% improvement over baseline).

PERSONAL PROJECTS

Personal AI Portfolio (anhhoangphucnguyen.com**)**
- Built an interactive Retrieval-Augmented Generation (RAG) application using LangChain, the Claude API, and VueJS to answer recruiter queries about my background.
- Engineered a vector database backend using Supabase (PostgreSQL with pgvector) to embed and retrieve context from personal documents.
- Architected CI/CID pipeline using Docker and GitHub Actions for delivery to Google Cloud Run.

SonicJob application - an employment platform to support both job seekers and recruiters
- Collected, cleaned and combined multiple job skill datasets to fine-tune RoBerta for Name Entity Recognition task, achieving F1 score of 81%.
- Built ClearML pipeline for data extraction, data preparation, model training, model evaluation, optimisation, and model selection.

PEER-REVIEWED PUBLICATIONS

- Evolving optimized transformer-hybrid systems for robust BCI signal processing using genetic algorithms https://doi.org/10.1016/J.BSPC.2025.107883
- EEG-TCNTransformer: A Temporal Convolutional Transformer for Motor Imagery Brain–Computer Interfaces.https://doi.org/10.3390/signals5030034 (Signals Best Paper Awards 2024)

CERTIFICATIONS

- Building with the Claude API | Claude Code in Action | Microsoft Certified: Azure AI Fundamentals